

AI Paper Roaster

Scientific Seminar Workflow

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Paper 1: Small Language Models are the Future of Agentic AI

Core Concept: AI should shift toward specialized, cost-effective small language models (under 10B parameters) rather than relying on massive cloud LLMs.

Skimming Questions to Answer

- What parameter threshold defines an "SLM", and what benchmarks proved competency?
- In what specific scenarios (tool use, planning, memory) do SLMs bottleneck?
- What is the operational cost and latency reduction ratio achieved?

The Weak Spot Assignment

Identify how the authors handle context window limitations, or find another weak spot.

Paper 2: Try, Check and Retry (TCR) Framework

Core Concept: The framework is called Tool-DC, and it uses a "Try-Check-Retry" paradigm to improve LLM tool-calling performance in long-context settings.

Skimming Questions to Answer

- How does the "Check" module work? Is it an LLM call or a rule-based parser?
- At what maximum context length does the TCR loop begin to degrade?
- How many "Retry" attempts does it take on average to fix a failed call?

The Weak Spot Assignment

Investigate compounding latency and token costs, or find another spot.

Paper 3: Physics of Agents

Core Concept: Adapting the Ising model from statistical mechanics to explain how large networks of LLM agents self-organize into indifference, polarization, or consensus.

Skimming Questions to Answer

- What exact physics model (e.g., Ising model) was adapted for agent interactions?
- What is the "critical temperature" tipping point parameter for polarization?
- Was this study purely theoretical or validated on real benchmarks?

The Weak Spot Assignment

Identify how the authors account for agent-level differences (personas, system prompts, model choice, seeds) — or critique their failure to do so.

Round 1: Breakout Rooms & Initial Findings

- **Action:** Choose your assigned paper's breakout room.
- **Time:** 25 Minutes to read and answer the 3 skimming questions, mention a weak spot.
- **Deliverable:** Make notes in a .doc, .md, or draft a short presentation.

7-Minute Presentation

Elect **one speaker** to focus on high-impact insights.

Ask **1 tough question** to another team (but no answers yet!)

Round 2: Problem Solving & Pitch Creation

Time Allocated: 20 Minutes back in Breakout Rooms

Task A: Formulate Defense

Draft a rigorous answer to the cross-examination question thrown at your team by the audience during Round 1.

Task B: Build the Pitch Deck (3–5 Slides)

- Elaborate on a novel architectural idea or solution framework.
- Explicitly present the **weakness you exposed** in your paper.
- Demonstrate how your new design fixes or bypasses that exact loophole.

Round 3: The Defend & Pitch Showdown

Phase	Execution Rule
1. The Defense	Directly answer the question from the audience first.
2. The Pitch	You have exactly 3 minutes for your slide presentation.

Timer is Strict. Structure for Maximum Impact.